www.tntextbooks.in

is faster than the sound we hear through the air (gaseous medium). This implies the velocity of sound is different in different media.

In this section, we shall derive the velocity of waves in two different cases:

1. The velocity of a transverse waves along a stretched string.
2. The velocity of a longitudinal waves in an elastic medium.

### 11.3.1 Velocity of transverse waves in a stretched string

Let us compute the velocity of transverse travelling waves on a string. When a jerk is given at one end (left end) of the rope, the wave pulses move towards right end with a velocity $v$ as shown in the Figure 11.15 (a). This means that the pulses move with a velocity $v$ with respect to an observer who is at rest frame. Suppose an observer also moves with same velocity $v$ in the direction of motion of the wave pulse, then that observer will notice that the wave pulse is stationary and the rope is moving with pulse with the same velocity $v$.

Consider an elemental segment in the string as shown in the Figure 11.15 (b). Let A and B be two points on the string at an instant of time. Let $dl$ and $dm$ be the length and mass of the elemental string, respectively. By definition, linear mass density, $\mu$ is

$$\mu = \frac{dm}{dl} \tag{11.5}$$

$$dm = \mu\, dl \tag{11.6}$$

The elemental string AB has a curvature which looks like an arc of a circle with centre at O, radius R and the arc subtending an angle $\theta$ at the origin O as shown in Figure 11.15(b). The angle $\theta$ can be written in terms of arc length and radius as $\theta = \frac{dl}{R}$. The centripetal acceleration supplied by the tension in the string is

$$a_{cp} = \frac{v^2}{R} \tag{11.7}$$

Then, centripetal force can be obtained when mass of the string (dm) is included in equation (11.7)

$$F_{cp} = \frac{(dm)v^2}{R} \tag{11.8}$$

The centripetal force experienced by elemental string can be calculated by substituting equation (11.6) in equation (11.8) we get

$$\frac{(\mathrm{dm})v^2}{R} = \frac{\mu v^2\, \mathrm{dl}}{R} \tag{11.9}$$

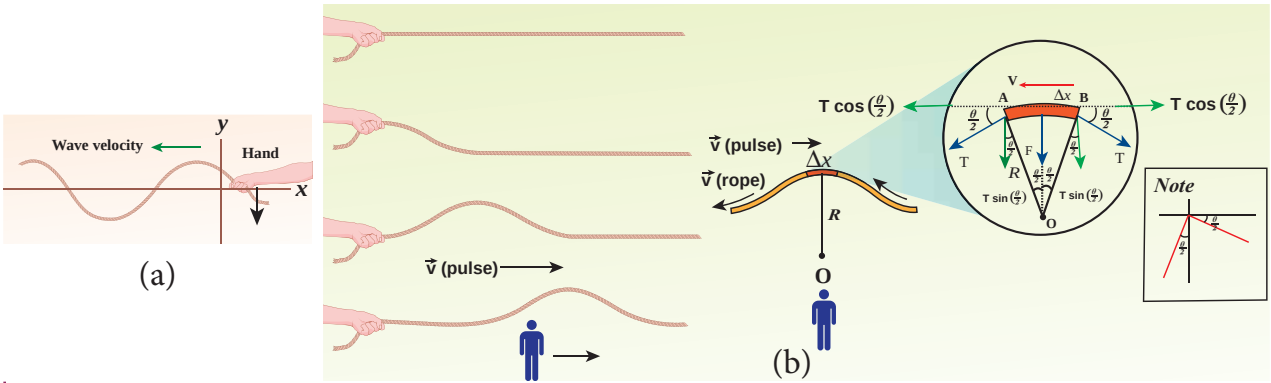



**Figure 11.15** (a) Transverse waves in a stretched string. (b) Elemental segment in a stretched string is zoomed and the pulse seen from an observer frame who moves with velocity $v$.